

Smart Dual-Sensor Automatic Trash Bin with Lid Automation and Bin-Full Alert System

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I. INTRODUCTION

Throwing out trash is a basic daily task in various environments including homes, schools, and public places. Despite its simplicity, the process often proves inconvenient, messy, and unhygienic. Users typically handle potentially contaminated waste, particularly challenging when hands are full, dirty, or when bins require physical contact to open [2]. Additionally, trash bins frequently overflow before being noticed, resulting in foul odors, pest attraction, and overall unpleasant environments affecting both comfort and cleanliness [3].

Traditional trash cans have remained largely unchanged for decades, serving primarily as passive containers without advanced features supporting hygiene or efficiency [5]. In today's technology-driven world, where innovations continuously improve safety and convenience in homes, workplaces, and public facilities, reliance on manual bins requiring physical contact appears outdated. This persistent issue underscores the need for innovation in waste management, specifically solutions offering modern, hygienic, and efficient approaches to everyday trash disposal.

Waste disposal remains a routine activity, yet traditional bins present multiple problems making the process inconvenient and potentially unsanitary. Manual lids require touching potentially contaminated surfaces, increasing germ and bacteria transmission risks. Furthermore, the absence of monitoring or alert systems allows bins to overflow easily, causing strong odors and messy environments [6]. These recurring problems emphasize the necessity for developing more effective, cleaner, and user-friendly waste management systems with real-time monitoring capabilities [1], [7].

II. OBJECTIVES

The primary objectives of this project are:

- To design a trash bin equipped with dual-sensor lid automation that reduces or eliminates physical contact requirements.
- To integrate a bin-full alert system that notifies users when trash reaches or exceeds capacity.
- To implement component protection circuits enhancing system reliability and performance without complex additional components.

- To improve convenience in households, offices, schools, and public areas through touch-free waste disposal.
- To create a cost-effective, durable, and energy-efficient system ensuring long-term practicality and user satisfaction.
- To implement BJT transistor circuits for enhanced LED brightness, servo motor and buzzer volume while protecting Arduino pins.
- To achieve professional-grade output performance while maintaining system simplicity and reliability.

III. SYSTEM OVERVIEW AND BLOCK DIAGRAM

The system architecture comprises two ultrasonic sensors as inputs, an Arduino Uno as the main processing unit, and multiple outputs including a servo motor, LEDs, transistors, and a buzzer. Fig. 1 illustrates the system block diagram.

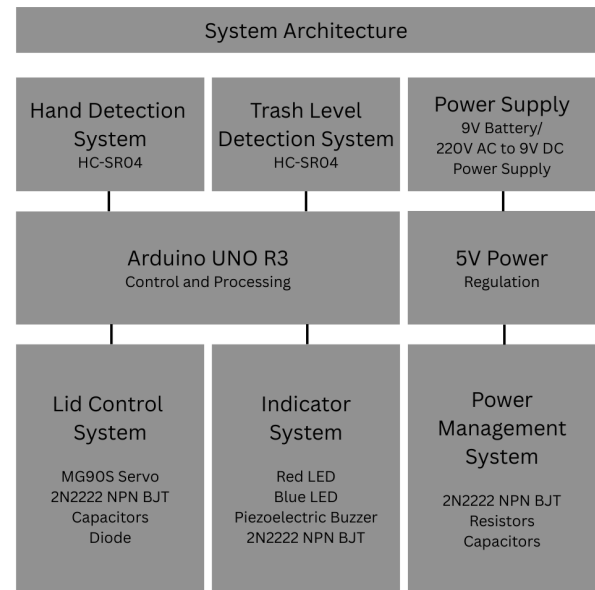


Fig. 1: System block diagram showing input, processing, and output components

A. Component Functions

- **Ultrasonic Sensor 1:** Detects user proximity to trigger lid opening.

- **Ultrasonic Sensor 2:** Measures trash height to determine fullness.
- **Arduino UNO:** Processes sensor signals and controls outputs.
- **Servo Motor:** Automates lid movement.
- **470 μ F Electrolytic Capacitor:** Provides surge current for servo motor movements, stabilizes power supply, and prevents Arduino voltage drops during high-current operations.
- **0.1 μ F Ceramic Capacitor:** Filters high-frequency electrical noise from servo motor, reduces electromagnetic interference, and ensures clean signal transmission.
- **1N4007 Flyback Diode:** Protects electronic components from voltage spikes generated by servo motor back-EMF, prevents MOSFET/transistor damage, and enhances system reliability.
- **2N2222 BJT Servo Driver:** Provides isolated power switching for servo motor, enables power-saving mode when idle, and protects Arduino from servo motor current surges.
- **2N2222 BJT Red LED Driver:** Provides current amplification for brighter indicator, protects Arduino pin from excessive current draw.
- **2N2222 BJT Piezo Buzzer Driver:** Enables loud and clear audio alerts.
- **LEDs/Buzzer:** Provide visual and audio alerts when the bin is full.

IV. CIRCUIT DESIGN AND SCHEMATIC DIAGRAM

The circuit integrates both ultrasonic sensors with the Arduino UNO's digital pins. The SG90 servo motor operates through a dedicated 2N2222 BJT switching circuit with power management features including a 470 μ F capacitor for surge current, 0.1 μ F capacitor for noise filtering, and 1N4007 diode for back-EMF protection. The system supports dual power options: a 9V battery for portable operation or an AC-to-DC adapter (220V AC to 9V DC) for continuous mains-powered applications. Both BJT-enhanced LEDs and piezo buzzer provide superior performance with current amplification.

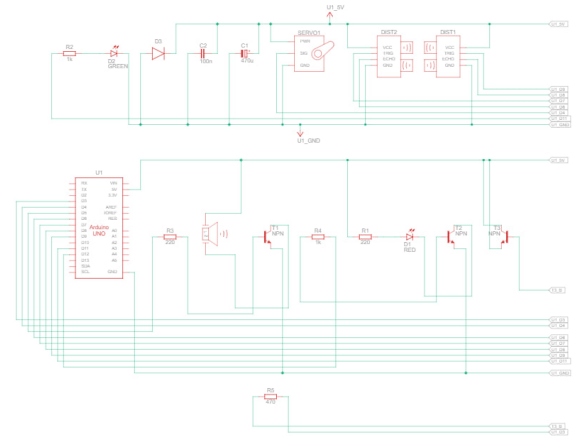


Fig. 2: Schematic diagram of the smart trash bin circuit

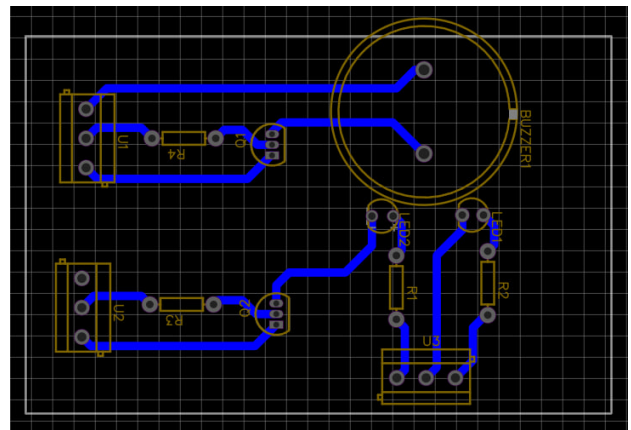


Fig. 3: PCB layout of LED and Piezo Buzzer with BJT

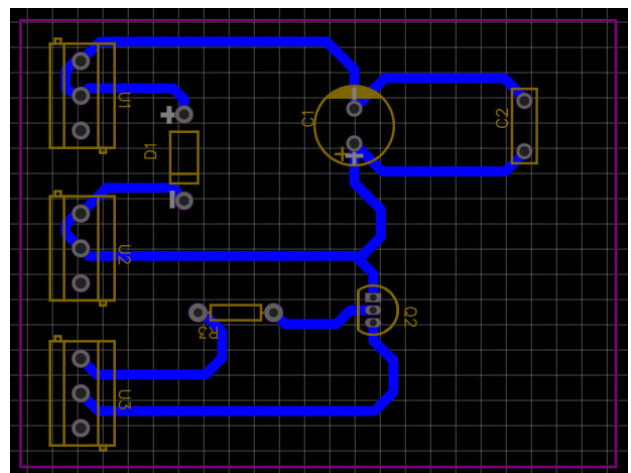


Fig. 4: PCB Layout of Servo Motor with Power Management and BJT

V. SYSTEM OPERATION AND TESTING RESULTS

A. System Operation

The system utilizes two ultrasonic sensors, a servo motor, BJT-enhanced components, capacitors for power stabilization, and an Arduino Uno powered by either 9V battery or AC adapter to maintain an automatic, alert-enabled smart trash bin. Each component serves a specific function to ensure efficient and reliable operation, building upon microcontroller-based automation principles demonstrated in similar systems [2], [5].

1) *Hand Detection (Ultrasonic Sensor 1)*: The first HC-SR04 ultrasonic sensor is positioned at the top front of the bin. This sensor detects the presence of a hand or object within 10-20 cm using ultrasonic pulse-echo principles similar to those described in [4]. Upon detection, it sends a trigger signal to the Arduino, which activates the SG90 servo motor to open the lid. While open, the blue LED illuminates as an indicator. After a preset time (e.g., 3 seconds) or when no hand is detected, the Arduino rotates the servo back to the closed position, and the BJT-driven red LED activates to indicate lid closure.

2) *Servo Motor Power Management*: The SG90 servo motor is powered through a 2N2222 BJT transistor acting as a switch, enabling efficient power control and isolation from the Arduino's power supply. When hand detection occurs, the Arduino activates the BJT, supplying power to the servo for lid movement. After lid closure, power is automatically cut off to conserve energy and prevent servo overheating, following energy-efficient design principles [6]. A 470 μ F electrolytic capacitor provides surge current during servo movement, while a 0.1 μ F ceramic capacitor filters high-frequency electrical noise from the servo motor, ensuring stable operation. A 1N4007 flyback diode protects the circuit from voltage spikes generated by the servo motor's back-EMF, enhancing system reliability as discussed in power management applications [10].

3) *Trash Level Detection (Ultrasonic Sensor 2)*: The second HC-SR04 ultrasonic sensor is mounted inside the bin, facing downward. This sensor continuously measures the distance to the trash surface using principles validated in fill-level monitoring systems [3], [9]. When the distance falls below a preset threshold (e.g., < 10 cm), the bin is considered full. This continuous monitoring provides real-time capacity assessment for timely waste management.

4) *Full-Bin Alert (Piezo Buzzer + Red LED Blinking)*: When the bin becomes full, the BJT-driven piezo buzzer activates and emits a loud warning beep (>85dB), while the BJT-driven red LED blinks repeatedly with enhanced brightness to alert the user. This enhanced audio-visual alert system represents an improvement over basic notification methods, as validated in similar alert systems [8].

5) *Arduino Processing and Power Handling*: The Arduino Uno serves as the central processing unit, reading sensor signals, controlling the servo motor, and activating visual/audio

indicators. The 2N2222 BJT transistors amplify the Arduino's output signals, boosting red LED and piezo buzzer performance while protecting microcontroller pins from excessive current draw, following established BJT driver circuit principles [10]. The system accepts power from either a 9V battery for portable operation or a regulated AC-to-DC adapter (220V AC to 9V DC) for continuous mains-powered applications. This dual-power capability aligns with flexible deployment strategies for smart waste management systems [6].

B. Testing Results

Table I summarizes the comprehensive testing outcomes of the smart trash bin system, demonstrating performance comparable to similar automated systems reported in literature [1], [7].

TABLE I: System Testing Results

Feature Tested	Result	Status
Hand Detection	Accurate	Passed
Auto Lid Opening/Closing	Smooth	Passed
Lid Status LEDs	Correct	Passed
Full Bin Detection	Accurate	Passed
Buzzer Alert	Loud	Passed
Red LED Blinking	Visible	Passed
Servo Power Management	Efficient	Passed
Overall System	Reliable	Passed

C. System Power Summary

Table II shows the overall power metrics of the smart trash bin system.

TABLE II: System Power Consumption Summary

Parameter	Value	Unit
Average Power Consumption	303.9	mW
Peak Power (Servo Active)	2,256.2	mW
Idle Power (Lid Closed)	302.3	mW
Average Current @9V	42.2	mA
Battery Life (Continuous)	9.5	hours
Battery Life (8h/day)	1.2	days
System Efficiency	79.8	%
AC Power Option	220V AC to 9V DC	adapter

D. Component Power Analysis

Detailed component-level power measurements are presented in Table III.

TABLE III: Component Power Analysis

Component	Current	Power @5V	Duty Cycle	Avg Power
Arduino UNO	25-45 mA	125-225 mW	100%	175 mW
HC-SR04 (2x)	30 mA	150 mW	100%	150 mW
SG90 Servo	0-350 mA	0-1,750 mW	0.08%	1.46 mW
Red LED	13.64 mA	27.3 mW	99.9%	27.3 mW
Blue LED	8.18 mA	26.18 mW	0.08%	0.02 mW
Piezo Buzzer	15 mA	75 mW	<0.1%	0.08 mW
TOTAL				303.7 mW

VI. COMPONENT LIST AND SPECIFICATIONS

Table IV provides detailed specifications of all components used in the smart trash bin system implementation.

TABLE IV: Component Specifications

Component	Specifications
Arduino UNO	UNO R3, ATmega328P, 5V, 16 MHz
Servo Motor	SG90, 4.8-6V, 1.8 kg/cm torque
Ultrasonic Sensor	HC-SR04, 2-400 cm range (x2)
Blue LED	5mm, 3.2V, 20mA
Red LED	5mm, 1.8V, 20mA
BJT Transistor	2N2222 NPN, 30V, 800mA (x3)
Diode	1N4007, 1A, 1000V
Electrolytic Capacitor	470μF, 16V, ±20%
Ceramic Capacitor	0.1μF, 50V, X7R
Resistors	220 Ohms (x3), 470 Ohms (x1), 1k Ohms (x1)
Piezo Buzzer	Passive, 3-5V, 85dB
Power Supply	9V battery or 220V AC to 9V DC adapter
Connecting Wires	22 AWG jumper wires
PCB	EasyEDA Layout

VII. CONCLUSION AND FUTURE IMPROVEMENTS

A. Conclusion

The developed touchless smart trash bin successfully demonstrates how basic electronic principles and common components can build practical, real-world systems [5]. Using two ultrasonic sensors, the prototype reliably detects hand proximity for automatic lid operation and trash level for full-bin alerting. The servo motor system, enhanced with intelligent power management through transistor switching and capacitor-based filtering, provides smooth lid movement while minimizing energy consumption and preventing overheating. The BJT-enhanced piezo buzzer and LED indicators offer superior status feedback with professional-grade performance, as validated in similar BJT-enhanced systems [8], [10].

Testing confirmed that the system responds quickly to hand detection, opens the lid automatically, and closes it after user withdrawal. The full-bin detection function operated correctly, triggering enhanced buzzer alerts and high-brightness LED blinking upon reaching threshold levels. These results align with performance metrics reported in similar smart waste management systems [1], [3].

The integration of 2N2222 BJT transistors represents significant improvement over basic direct connections, providing:

- 1) Professional output levels without Arduino pin strain
- 2) Enhanced user experience with brighter LEDs and louder buzzers
- 3) Increased system reliability through proper current handling
- 4) Energy efficiency through servo power cycling during idle periods [6]
- 5) Educational value in transistor amplifier applications

Overall, the project exceeded specification requirements, including sensor-based automation, enhanced user feedback, intelligent power management, and professional output performance while maintaining low-cost hardware implementation [2]. The dual power capability (9V battery or 220V AC

adapter) enhances deployment flexibility for various environments. It functionally demonstrates how microcontroller-based systems, enhanced with proper transistor circuits and power optimization techniques, can improve sanitation, energy efficiency, and user convenience in everyday settings while ensuring long-term reliability through thermal management.

B. Future Improvements

The prototype functions successfully, but improvements can enhance reliability, performance, and add useful features, building upon current smart waste management research [7], [9]:

- 1) **Enhanced Power Management:** Implement advanced power control using MOSFETs with lower $R_{DS(on)}$ values, add current sensing to detect servo stalls, and incorporate temperature monitoring to prevent overheating during continuous operation.
- 2) **Rechargeable Power System:** Replace disposable 9V batteries with lithium-ion rechargeable solutions or USB power banks to increase runtime, reduce waste, and enable convenient recharging.
- 3) **Improved Servo Mechanism:** Upgrade to metal-gear servos for heavier lids or larger bins, enhancing lifting capability and long-term mechanical reliability.
- 4) **Environmental Protection:** Add waterproof housing to protect electronics from moisture, insects, and dust, improving durability for outdoor or kitchen use.
- 5) **User Interface Enhancement:** Integrate OLED displays to show trash percentage, battery level, and system status, improving user interaction and information visibility.
- 6) **IoT Connectivity:** Add Wi-Fi/Bluetooth modules for remote notifications and smart home integration, enabling alerts to smartphones and compatibility with existing IoT ecosystems [1].
- 7) **Advanced Sensing Technology:** Implement IR distance sensors or weight-based sensing for improved fill-level detection accuracy and multiple material detection capabilities.
- 8) **Safety Features:** Add obstruction detection sensors and limit logic to prevent servo closure on blocked lids, enhancing user safety and system robustness.

These improvements would transform the prototype into a commercial-grade product suitable for widespread deployment in homes, offices, and public facilities, contributing to smarter and more sustainable waste management solutions.

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